

Question:

Elly is organizing her books on the new bookcases her parents bought her. Each of the middle 2 shelves can hold 10 books. The bottom shelf can hold twice as many books as a middle shelf. The top shelf can hold 5 fewer books than the bottom shelf. If she has 110 books, how many bookcases does she need to hold all of them?

Reasoning + Answer:

<think>The bottom shelf can hold twice as many books as the middle shelf, or a total of $20 * 2 = 40$ books. The top shelf can hold 5 fewer books than the bottom shelf, or a total of $40 - 5 = 35$ books. The three middle shelves can hold a total of $20 + 40 + 35 = 95$ books. This means that each bookcase can hold a total of 95 books, meaning she will need $110 / 95 = 1.15$ books per bookshelf. Since she can't have a fraction of a bookcase, she will need 2 bookcases.
</think><answer>2</answer>

Reward score

300

200

100

0

-100